

Causal Value Models for Adaptive Prompting

Theory, statistical reference, and experimental-agent protocol

DTR-MultiRoundLLM research development package

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This package develops supported history-conditional prompt effects and policy value for repeated interaction with a fixed receiver model. It includes mathematical proofs, a reproducible synthetic statistical reference, and a concrete protocol for the next experimental agent. Real-model efficacy, trained neural calibration, and unrestricted prompt optimality remain unestablished.

The maintained repository, complete literature audit, proof-review record, code, raw synthetic results, and work-package issues are available at [DTR-MultiRoundLLM](#). This PDF is a dated development snapshot.

Causal value models for adaptive prompting

Scientific question

For a new task and a fixed receiver model, which next prompt or feedback intervention improves the final task outcome, given the conversation so far and a specified future interaction policy? Can a model learn these conditional values, use them to rank newly generated prompts, and eventually serve as an autonomous adaptive user?

The initial request defines the task. An initial receiver answer is observed before the first feedback decision. Prompt rewriting before the initial answer is a separate, optional decision at time zero. Neither formulation identifies the effect of the task’s semantic content merely by comparing unrelated requests.

The proposed system has four components: a history encoder; a prompt generator that proposes exact natural-language interventions; a continuation-specific causal value model; and a selector that decides among candidates and STOP. The receiver weights, tools, system instruction, decoding, and context rules are frozen within a study. The first paper focuses on prompt-only feedback. Tool availability, model switching, and external information provision change the intervention and are extensions.

What a personalized prediction means

For history h , generated slate c , candidate j , and continuation policy π , the model estimates $Q_t^\pi(h, c, j)$, the expected final utility if that candidate is selected now and π governs later choices. Contrasts between two candidates give a history-conditional causal effect. The prediction averages over future receiver randomness and relevant unobserved variation; it is not the realized counterfactual effect for one person or one exact trajectory.

Three outputs should be displayed separately:

Output	Meaning	Validation
Predicted final quality under candidate j	A conditional expected outcome with a named continuation policy	Fresh branch outcomes, calibration, prediction error
Difference from a reference candidate or STOP	Expected improvement for comparable supported histories	Randomized contrasts; repeated branches estimate conditional means
Estimated policy value	Average utility of the complete adaptive strategy in a target task population	Independent prospective trial and task-cluster uncertainty

For an unseen history the prediction is a generalization claim. For an unsupported prompt it is extrapolation. An uncertainty estimate for average policy value does not supply a valid interval for every prompt in every history. The first system reports support flags and calibration bins, with explicit abstention when support is poor; it does not attach unproved pointwise causal confidence intervals.

Contribution and its boundary

The contribution under development is a **design and estimation framework for sequential language-valued interventions**: generate a slate, randomize the actual intervention, retain the complete assignment mechanism, estimate continuation-specific values, and evaluate an adaptive prompting policy prospectively. The slate design allows free-form text candidates without pretending that every possible string has usable observational support. It also makes clear when changing the generator changes the target distribution.

DTR identification, g-computation, longitudinal doubly robust estimation, actor–critic optimization, and automatic iterative feedback are established foundations. Correctly specified sequential outcome regression can identify these targets under the same causal assumptions; a conventional critic is not automatically noncausal. Orthogonal or doubly robust learning protects against specified nuisance errors, not unmeasured confounding, omitted history, arbitrary model shift, or an unsupported action space. The literature audit records which closely related methods already exist.

The intended methodological claims are (i) an explicit language-action estimand and experimentally identifiable design; (ii) proofs connecting selection-only versus joint generator/selection policy evaluation; (iii) error decompositions separating candidate availability from ranking and continuation error; and (iv) a calibrated empirical comparison of value learning and autonomous feedback. The current package does not establish that this combination is unprecedented, that a new neural architecture is statistically optimal, or that it improves real LLM outcomes.

From evaluation to a trainable dynamic skill

1. Collect randomized multi-round interaction data from frozen candidate generators, preserving histories, all candidates, selection probabilities, STOP, and objective outcomes.
2. Train a history–prompt critic using backward fitted evaluation and conditional orthogonal pseudo-outcomes. Name the continuation policy in every checkpoint. Use task-level cross-fitting and a final held-out evaluation set.
3. Rank generated candidates by estimated net downstream utility; apply a support screen and a separately calibrated improvement threshold over a reference policy.
4. Train a prompt generator from high-value candidates or a score-function objective on a frozen critic. This is a proposal mechanism; it can exploit critic error and needs fresh randomized data after updates.
5. Freeze the generator and selector, then compare full adaptive trajectories against cost-matched baselines on new tasks. Iterate training only between declared collection batches.

An initial implementation can use a frozen text encoder and small critic head. A sequence model and generator fine-tuning are subsequent implementations of the same contract. A learned world model is optional and is not substituted for external validation. The dynamic skill is the versioned mapping from history to intervention, with stopping and cost accounting, rather than a fixed sequence of textual instructions.

Outcomes and application choice

The primary scientific outcome is independently verifiable final task success. Secondary outcomes are terminal quality on a fixed rubric, regressions from correct to incorrect, total receiver/generator/critic tokens, wall-clock time, and intervention count. Report the quality–cost frontier; a scalar utility is useful for policy training only after a cost conversion is declared. Added information and oracle error labels form separate, explicitly privileged treatment arms.

The first real application should be a competent open-weight model on verifiable reasoning or constrained structured-output tasks, followed by isolated code-repair tasks if a secure runner is available. Synthetic arithmetic is a harness check, not the substantive application. The experimental protocol requires a pilot to avoid both floor and ceiling performance before freezing a benchmark.

Relation to the earlier projects

The earlier routing project intervenes on the agent’s internal model/tool choice. This project intervenes on what the user or virtual user says next while the receiver is fixed. Prioritized multivariate outcome comparisons can be added later, but their pairwise sampling and estimand require their own analysis. The first version preserves correctness and cost as separate endpoints and does not import a win-ratio estimator without a derivation.

Evidence required for a paper

The proof document and reference simulations support mathematical and implementation checks. A credible empirical paper also needs fitted-critic validation, repeated-prefix branch calibration, a generator-update experiment, and an independent cost-matched policy trial. Until those exist, this is a theory and experiment development package, not a submission-ready empirical manuscript.

Causal value models for adaptive prompting: estimands, identification, estimation, and decisions

Version: 2026-09-19. **Status:** results proved below under the stated assumptions; initial mathematical audit completed, independent review pending. This is an initial research theory package, not a claim of new foundational DTR theory or completed empirical validation. Companion audit: [theory_proof_audit.md](#).

1. Scientific question and contribution boundary

For a fixed receiver LLM, tool environment, task distribution, and evaluator, what would happen if the next feedback prompt were changed, and which supported adaptive prompting policy improves final task performance after accounting for cost? The intended deployed system observes a request and the receiver’s answer, generates possible feedback, scores their downstream values, chooses feedback or STOP, and repeats. We call the score a **causal prompt value model** only when the intervention, continuation policy, outcome, and identification assumptions are explicit.

The object is a history-conditional expected counterfactual outcome or contrast, not an individual’s unknowable realized treatment effect. A neural value predictor and a generative actor do not themselves establish causality. Randomization or defensible sequential exchangeability establishes identification; function approximation estimates an identified functional; independent evaluation establishes whether decisions improve.

The core DTR, g-computation, importance weighting, augmentation, and policy improvement arguments are established statistical ideas. This package specializes and connects them to a two-stage language intervention: **generate a candidate slate, then randomize a selected candidate**. Potential research contributions are the support-aware language intervention design, history-conditional learning and calibration, candidate-generation versus selection error decomposition, and empirical evidence that the resulting decisions improve over adequately adjusted alternatives. Novelty requires a separate literature audit. Merely using a Transformer, calling a critic causal, or applying DTR to prompts is insufficient.

2. Timeline, unit, and exact intervention

2.1 Unit and time origin

The primary independent unit is a **root task** sampled from a declared target population. A root can produce several randomized continuations, but those continuations are not independent roots. A user with several related tasks can instead be the independence cluster when cross-task dependence is material.

Let X be the fixed original user request and baseline metadata; let O_0 be the receiver’s initial answer. Set $H_1 = (X, O_0, \text{baseline metadata})$. Turns $t = 1, \dots, T$ are subsequent feedback opportunities. Thus this version estimates feedback effects conditional on the initiating request; it does not estimate the effect of choosing an entirely different original task. To study initial-prompt wording, hold the underlying task and target answer fixed and add an initial randomized wording decision before O_0 .

At turn t :

1. Observe full pre-generation history H_t .
2. Draw an ordered candidate slate $C_t = (a_{t1}, \dots, a_{tK_t})$ from a frozen generator kernel $G_{0t}(dc|H_t)$.
3. Observe all eligible candidates, including an explicit STOP candidate, then draw index J_t from logging selector $b_t(j|H_t, C_t)$.
4. Render and send the exact selected intervention $A_t = a_{tJ_t}$; observe response, costs, and the next history.

The mathematically general action is $U_t = (C_t, J_t)$. Treating this compound action as the intervention avoids hidden assumptions about whether unselected candidates affect cost or future controller behavior. A contrast between j and j' within the same realized slate holds generation fixed and changes the selected prompt.

H_t includes every variable used by the controller or generator, receiver-visible transcript, prior actions, past outputs, available tool results, remaining budget, model versions, and relevant logged prior slate/selection information. It never includes a hidden test label or future evaluator result unavailable at deployment. If previous candidates are neither retained nor used by

any future mechanism, they can be omitted by a declared conditional-independence restriction. A learned embedding is not automatically a sufficient replacement for H_t .

Candidate order, duplicates, truncation, formatting, tool attachments, forbidden-action filters, and rendering are part of treatment definition. The same text at two indices can have different index probabilities. Keep indices and the full slate in the log; do not silently treat index propensity as marginal text propensity.

2.2 Frozen environment and rewards

Freeze receiver model/version, system instruction, decoding parameters, tools, retrieval corpus snapshot where feasible, and evaluator definition. Stochastic receiver responses remain part of the transition distribution. New model versions or externally changing tool state require a new environment stratum or a transport assumption, not a silent pooling operation.

Let R_t be a bounded turn utility, often minus the measured incremental computational cost. Let $Y = y(H_{T+1})$ be terminal quality under a prespecified scoring rule. The total utility is

$$U^{\text{tot}} = \sum_{t=1}^T R_t + Y.$$

One equivalent implementation sets all $R_t = 0$ and includes cumulative costs in terminal Y . Do not count costs twice. Report quality, cost, and stopping time separately even when optimizing $Y - \lambda \text{ cost}$; choose λ before the confirmatory comparison. Resource-constrained optimization and multivariate outcomes are valid alternative targets but are not automatically summarized by a single scalar value.

Generation cost is incurred before selection and is charged even if STOP is then chosen. A cheaper pre-generation stopping gate is a distinct decision stage. A baseline that receives fewer model calls is not cost-matched merely because both systems use the same number of feedback turns.

2.3 STOP, missingness, and failure

STOP accepts the current output. It is an observed treatment that leads to an absorbing state: after STOP, pad the horizon with a deterministic dummy STOP, no further cost, no new receiver call, and an unchanged terminal output. The deterministic padding has likelihood ratio one. Randomized support is required for the actual stop/continue decision, not for artificial padding.

An API failure, missing final score, or lost log is not STOP. A timeout can be a prespecified unfavorable outcome, or it can be censoring requiring a separate observation model. Deleting failed calls estimates a selected-success population. The theorems below assume all defined rewards and terminal outcomes are observed. Missing-at-random and positive observation probabilities can justify added censoring weights; informative missingness is not repaired by treatment weights alone.

3. Policies and estimands

A target policy consists of an externally specified generator G_t^* and selector π_t ; write $d_t(dc, j|h) = G_t^*(dc|h)\pi_t(j|h, c)$. The logging action kernel is $e_t(dc, j|h) = G_{0t}(dc|h)b_t(j|h, c)$. Integrals below include a sum over the finite slate indices.

Main first-stage deployment/evaluation freezes $G_t^* = G_{0t}$ and learns only the selector. Its **marginal text policy** is

$$\bar{\pi}_t(a | h) = \int G_{0t}(dc | h) \sum_j \pi_t(j | h, c) 1\{c_j = a\}.$$

The logged probability $b_t(J_t|H_t, C_t)$ is a conditional selection propensity, not $P(A_t = a|H_t)$ and not a candidate generator token probability.

Define remaining-utility value and action value by a policy-specific Bellman recursion:

$$V_{T+1}^d(h) = y(h), \quad Q_t^d(h, c, j) = E\{R_t + V_{t+1}^d(H_{t+1}) \mid H_t = h, \text{do}(C_t = c, J_t = j)\},$$

$$W_t^d(h, c) = \sum_j \pi_t(j | h, c) Q_t^d(h, c, j), \quad V_t^d(h) = \int G_t^*(dc | h) W_t^d(h, c), \quad \mathcal{V}(d) = E\{V_1^d(H_1)\}.$$

Every Q depends on the **future** policy. A score trained for generic retry continuation cannot be relabeled as the value under a newly optimized autonomous continuation without updating or reevaluating that continuation.

The within-slate history-specific contrast is

$$\tau_t^d(h, c; j, j') = Q_t^d(h, c, j) - Q_t^d(h, c, j').$$

This answers: among roots at this history and realized slate, what expected downstream utility changes when selecting this candidate instead of the comparator and then using d ? STOP, a specific fixed prompt, or a prespecified stochastic baseline can be the comparator. Historical costs before this decision cancel in the contrast.

A more compact $Q_t^d(h, a)$ requires a **slate-exclusion assumption**: conditional on h and exact rendered a , transition utility and every future policy-relevant variable are invariant to unselected candidates and index. If generation cost is already sunk and subtracted from both compared values, exclusion can hold for the remaining receiver outcome even when it fails for total system cost. If future policies retain slate content, or the receiver sees multiple candidates, exclusion can fail. In those cases retain (h, c, j) .

For a newly arriving user, these are population conditional means for histories like the observed history under the stated environment. Exact-history conditional means are defined almost surely; a unique long transcript typically has no replicated observations. Neural generalization and uncertainty at that transcript require regularity and training support beyond identifiability of a population functional.

4. Identification assumptions

A1 (well-defined intervention and consistency). Selected rendered prompt, slate-generation protocol, STOP, response mechanism, and utility are sufficiently specified. Observed outcomes equal the corresponding potential outcomes for the realized actions. There is no interference across independent root clusters; copying a live state and executing a branch must not mutate another branch’s tools or hidden memory.

A2 (sequential exchangeability). Conditional on full H_t , assignment of U_t is independent of future potential rewards and states under supported interventions. For fixed-generator selection comparisons this can be stated in two steps: slate generation uses only H_t and independent randomness; conditional on (H_t, C_t) , randomized J_t is independent of future potential outcomes. Observational human-feedback logs need all common causes of treatment and outcome measured; latent human knowledge or inaccessible task facts can violate this condition.

A3 (sequential support). At every history reachable under the target, d_t is absolutely continuous with respect to e_t . Ratios must be finite for identification; bounded or sufficiently integrable products are additionally needed for precision and asymptotic inference. Under the same generator the requirement reduces to $\pi_t(j|h, c) > 0 \Rightarrow b_t(j|h, c) > 0$, together with support along the preceding target trajectory. Deterministic target selection is permitted when its selected indices have logging support.

A4 (stable transition and measurement law). Conditional transition kernel $P_t(dr, dh'|h, c, j)$ and scoring rule apply both in logging and target execution. This is not assured by adjusting for a model-name string if the underlying model, tools, evaluator, or task distribution changed.

A5 (sampling and integrability). Root clusters are independent under the primary design; relevant moments exist. Additional boundedness/rate conditions are stated at the result where needed.

These assumptions do not require a low-dimensional Markov state: full history is always the state in this formulation. They also do not make observational confounding an inherent flaw of ordinary Q regression. Correct sequential regression on full sufficient history identifies the same causal Q under A1–A4. A fair comparison includes this adequately adjusted regression, as well as confounded marginal reward prediction.

Theorem 1: longitudinal identification and change of measure

Under A1–A4, the counterfactual law under d is identified by

$$p^d(dh_1, d\bar{u}, d\bar{r}, d\bar{h}') = p(dh_1) \prod_{t=1}^T d_t(du_t | h_t) P_t(dr_t, dh_{t+1} | h_t, u_t). \quad (1)$$

The preceding Bellman equations identify Q_t^d , V_t^d , and $\mathcal{V}(d)$ on supported histories. With

$$\rho_t = \frac{d_t(U_t | H_t)}{e_t(U_t | H_t)}, \quad \Omega_t = \prod_{s=1}^t \rho_s, \quad \Omega_0 = 1,$$

and integrable quantities, the value also satisfies

$$\mathcal{V}(d) = E_e \left[\sum_{t=1}^T \Omega_t R_t + \Omega_T Y \right]. \quad (2)$$

Proof. Consistency identifies the observed transition conditional on a realized treatment with the corresponding potential transition. Sequential exchangeability removes selection into that treatment conditional on prior full history. Apply these facts at the final decision, then induct backward to replace every counterfactual transition by P_t . Integrating the last transition proves the terminal Bellman step; iterating proves (1) and the complete recursion. Absolute continuity implies that the target law through transition t has Radon–Nikodym derivative Ω_t with respect to the logging law through that transition; all transition factors cancel. Therefore $E_d R_t = E_e(\Omega_t R_t)$ and $E_d Y = E_e(\Omega_T Y)$. Sum these equalities. No cross-world joint distribution of two different action outcomes has been used. $\square$

Same-generator corollary. If $G^* = G_0$, then $\rho_t = \pi_t(J_t | H_t, C_t) / b_t(J_t | H_t, C_t)$. The generator probabilities cancel even if a black-box generator cannot provide tractable sequence likelihoods. This does **not** authorize changing that generator while retaining the ratio.

Generator-change corollary. For a different generator with supported, evaluable density ratio,

$$\rho_t = \frac{dG_t^*(\cdot | H_t)}{dG_{0t}(\cdot | H_t)}(C_t) \frac{\pi_t(J_t | H_t, C_t)}{b_t(J_t | H_t, C_t)}. \quad (3)$$

This is the density of the complete ordered slate after its actual sampling, filtering, and deduplication procedure. Multiplying raw token probabilities is correct only if it exactly reproduces that procedure. Without support or a valid generator ratio, collect new randomized trajectories under the revised frozen generator. Updating an actor is an engineering operation; it is not automatically an identifiable offline causal evaluation.

Proposition 2: limits of individual and unsupported-prompt claims

- (a) Randomized data do not generally identify the distribution of literal individual effects $Y_i(a) - Y_i(a')$.
- (b) Without support or structural extrapolation assumptions, the value of an arbitrary unseen prompt is not nonparametrically identified.

Proof. For (a), let the two potential outcomes each be Bernoulli with probability one half. In model I they are identical; in model II they are complements. Independently randomized treatment yields the same observed distribution in both models. Every individual effect is zero in I, whereas effects are -1 or $+1$ in II. Thus the observed law identifies both means and their difference, but not the individual-effect distribution. For (b), choose an action with zero logging probability at a target-relevant history. Two transition kernels can agree on every observed action and assign terminal means zero and one, respectively, to this action. They induce the same logging law and different target values. $\square$

Formal positive token probability for every string is weaker than useful support: probabilities can be so small that finite-sample estimation is effectively unconstrained. Embedding proximity and a language-model prior are extrapolation assumptions, not replacements for this argument.

5. Representations, text treatments, and what a neural critic must learn

Let $S_t = f(H_t)$ and $Z_t = z(A_t)$ be representations. Three distinct claims should not be conflated:

- **Prediction approximation:** a sufficiently flexible $q(S, Z)$ predicts observed outcomes well.
- **Decision sufficiency for a fixed continuation:** $Q_t^d(h, c, j) = q_t^d(f(h), z(c_j))$ for every supported action under the declared exclusion restrictions.
- **Controlled Markov sufficiency:** conditional reward and the next represented state depend on full history and intervention only through the represented state/action; terminal quality is a function of the terminal representation; target/logging policies and generator laws are representable on the same state when a reduced-state likelihood is used.

The second permits estimating the specified Q; the third supports recursive planning and reusable Bellman equations for policies measurable in the representation. Neither follows from a high-dimensional encoder architecture or from predicting average reward accurately.

Proposition 3: sufficient representations and a counterexample

Suppose for all admissible actions the conditional law of (R_t, S_{t+1}) given full history and action depends only on (S_t, Z_t) , terminal Y is measurable in S_{T+1} , slate exclusion holds, and the target generator/selector uses only the represented variables. Then backward recursion gives $Q_t^d(h, a) = q_t^d(S_t, z(a))$.

Proof. At the terminal stage the value is measurable in S_{T+1} . If V_{t+1}^d is a function of S_{t+1} , conditional expectation of $R_t + V_{t+1}^d$ is a function of (S_t, Z_t) by the controlled transition restriction. Integrating over a represented target policy makes V_t^d a function of S_t . Induct backward. $\square$

Without these restrictions, let H be a fair binary variable, A be independently randomized binary treatment, and $Y = 1(A = H)$. An encoder $S = 0$ loses no marginal mean prediction beyond the constant one half, but both encoded action means equal one half; the full-history policy $A = H$ achieves one and an encoded policy achieves at most one half. Thus randomization survives in this example while action-relevant heterogeneity is destroyed. In observational data, collapsing a true confounder can additionally destroy exchangeability.

A prompt embedding $z(a)$ can collapse wording variants with different effects. A well-defined intervention $do(Z = z)$ then requires either treatment-version irrelevance or a specified distribution over exact strings mapping to z . The latter is a stochastic textual intervention, with value depending on that distribution. Decoding an optimized embedding into a new string does not prove either property.

6. Longitudinal doubly robust evaluation

6.1 Target fixed outside the evaluation sample

First freeze d independently of the evaluation sample. This can be a prespecified policy or a policy learned on separate training roots. All nuisance functions below are also fixed when taking conditional expectations; cross-fitting implements this condition fold by fold.

Let $\widehat{Q}_t(h, u)$ be any bounded candidate Q functions and let

$$\widehat{V}_t(h) = \int d_t(du | h) \widehat{Q}_t(h, u), \quad \widehat{V}_{T+1} = Y.$$

Let $\widehat{e}_t$ be positive on target support and define

$$\widehat{\rho}_t = d_t / \widehat{e}_t, \quad \widehat{\Omega}_t = \prod_{s=1}^t \widehat{\rho}_s.$$

The score is

$$D_d(O; \hat{\eta}) = \widehat{V}_1(H_1) + \sum_{t=1}^T \widehat{\Omega}_t \{R_t + \widehat{V}_{t+1}(H_{t+1}) - \widehat{Q}_t(H_t, U_t)\}. \quad (4)$$

A terminal-utility implementation has $R_t = 0$ and $\widehat{V}_{T+1} = Y$ including its accumulated costs. The cross-fitted value estimate averages (4), with every root and all of its branches confined to one fold.

Theorem 4: exact remainder and stagewise robustness

Let $\delta_t = \widehat{Q}_t - Q_t^d$, and assume integrability. Then

$$E_e D_d(O; \hat{\eta}) - \mathcal{V}(d) = \sum_{t=1}^T E_e \left[\widehat{\Omega}_{t-1} \int d_t(du \mid H_t) \left\{ 1 - \frac{e_t(u \mid H_t)}{\widehat{e}_t(u \mid H_t)} \right\} \delta_t(H_t, u) \right]. \quad (5)$$

Thus the score is unbiased when all assignment mechanisms are correct or all target-continuation Q functions are correct. More generally, each stage's summand is zero if **at that stage** either $\widehat{e}_t = e_t$ or $\widehat{Q}_t = Q_t^d$. Arbitrary stagewise choices of these conditions suffice. This statement concerns actual target Q functions, not regression fits that may inherit misspecification from earlier pseudo-outcomes.

Proof. Write $\widehat{V}_t = V_t^d + \delta V_t$ where $\delta V_t(h) = \int d_t \delta_t$, and $\delta V_{T+1} = 0$. For every t , the true Bellman residual $R_t + V_{t+1}^d - Q_t^d$ has conditional mean zero given (H_t, U_t) , hence its product with $\widehat{\Omega}_t$ also has mean zero. Subtract $\mathcal{V}(d)$ from the mean of (4). The remaining expression is

$$E \delta V_1 + \sum_{t=1}^T E \{ \widehat{\Omega}_t \delta V_{t+1} - \widehat{\Omega}_t \delta_t(H_t, U_t) \}.$$

Reindex the first term of the sum, using $\widehat{\Omega}_0 = 1$ and $\delta V_{T+1} = 0$, to obtain

$$\sum_{t=1}^T E \{ \widehat{\Omega}_{t-1} \delta V_t - \widehat{\Omega}_t \delta_t(H_t, U_t) \}.$$

Conditional on H_t , integrate the second term under the true action law e_t : it equals $\widehat{\Omega}_{t-1} \int e_t(d_t/\widehat{e}_t) \delta_t$. Subtraction gives (5). All robustness statements follow by making either factor in each integrand zero. $\square$

For a fixed same generator, use $\widehat{e} = G_0 \widehat{b}$ and $d = G_0 \pi$. Equation (5) reduces to

$$\sum_t E \left[\widehat{\Omega}_{t-1} \int G_{0t}(dc \mid H_t) \sum_j \pi_t(j \mid H_t, c) (1 - b_t/\widehat{b}_t) \delta_t(H_t, c, j) \right]. \quad (6)$$

Known randomization probabilities should ordinarily be used as logged. Estimated propensities cannot fix missing intervention support. Weight clipping changes the exact identity and trades bias for variance; report it as a sensitivity estimator alongside unclipped identification targets.

6.2 Orthogonality and rates

Equation (5) is a product remainder. At truth, a small perturbation in either nuisance while the other stays true has zero first derivative in the score mean. This is the claimed Neyman orthogonality; it is not a guarantee about a neural optimizer finding the correct functions.

For a conditional-on-training statement, define the finite weighted stage measure

$$\mu_t(dh, du) = E_e\{\widehat{\Omega}_{t-1}1(H_t \in dh)\} d_t(du | h).$$

Cauchy–Schwarz applied directly to (5) gives the precise bound

$$|ED_d - \mathcal{V}(d)| \leq \sum_{t=1}^T \left\| 1 - \frac{e_t}{\widehat{e}_t} \right\|_{L_2(\mu_t)} \|\widehat{Q}_t - Q_t^d\|_{L_2(\mu_t)}. \quad (7)$$

The assignment error is **relative** to the fitted assignment density. Bounded likelihood ratios alone do not convert it into an ordinary absolute density-error norm when densities can approach zero. In the fixed-generator finite-selection design, the explicit lower bound $\widehat{b}_t(j|h, c) \geq \varepsilon > 0$ gives $|1 - b_t/\widehat{b}_t| \leq |b_t - \widehat{b}_t|/\varepsilon$; dominated weighted stage measures then give the familiar absolute-propensity-error product bound. Analogous absolute-density claims for unrestricted language require an appropriate lower-bound/domination condition and should not be assumed.

For fixed T , products in (7) that sum to $o_p(n^{-1/2})$, score convergence in L_2 , and independent-root cross-fitting yield asymptotic linearity. Both the relative assignment error and the Q error converging at $o_p(n^{-1/4})$ in these weighted norms is one sufficient regime, not a necessary universal requirement. The mass and domination constants of these measures can grow exponentially with horizon; the bound does not remove the curse of overlap. With known randomized e , the bias remainder is exactly zero even for misspecified Q, although efficiency then depends on Q quality.

Asymptotic expansion. With a fixed finite number of folds, condition on a fold’s training sample. Decompose its score average into the empirical mean of the true score, the centered empirical mean of $D(\widehat{\eta}) - D(\eta_0)$, and its conditional mean. Conditional independence and L_2 score convergence make the centered difference $o_p(n^{-1/2})$ by its conditional variance; (7) controls the mean. Sum the finite folds. The ordinary independent-root central limit theorem then applies to the true influence scores. This proof fails if branches from one root leak across training and evaluation folds.

Theorem 5: efficient influence function in the stated model

Consider a nonparametric finite-horizon full-history model whose unknown factors are the initial-history law, logging action laws, and transition/reward laws. The target action law d is fixed externally and does not vary with this observed-data model. Assume positivity, differentiability along regular submodels, and square-integrability of the expression. Then

$$\phi_d(O) = V_1^d(H_1) - \mathcal{V}(d) + \sum_{t=1}^T \Omega_t \{R_t + V_{t+1}^d(H_{t+1}) - Q_t^d(H_t, U_t)\} \quad (8)$$

is the canonical gradient, hence efficient influence function. It remains canonical if known logging randomization mechanisms are removed from this model.

Proof. A regular submodel score decomposes into $S_0(H_1) + \sum_t S_{e,t}(U_t|H_t) + \sum_t S_{P,t}(R_t, H_{t+1}|H_t, U_t)$, each component having conditional mean zero at its factor. Differentiating the g-formula, the initial-law derivative is $E[(V_1^d - \mathcal{V})S_0]$. The target law does not contain e , so every logging-law derivative is zero. For a transition perturbation at stage t , its derivative is the target expectation of remaining return times $S_{P,t}$. Conditional expectation of the later return is $R_t + V_{t+1}^d$; earlier rewards are measurable before this transition and contribute zero. Change of measure and subtraction of the conditional mean yield $E[\Omega_t(R_t + V_{t+1}^d - Q_t^d)S_{P,t}]$.

Now pair (8) with the whole score. Conditional mean-zero and iterated expectation eliminate all cross-factor terms; only the displayed derivative for the matching factor remains. Also $E\phi_d = 0$. Each summand is a permissible initial or transition tangent component multiplied by a function measurable before its transition. Thus (8) belongs to the tangent-space closure and represents every pathwise derivative, which makes it the canonical gradient. It is orthogonal to all logging-action tangent components; removing known-action score components leaves the same canonical gradient. $\square$

Model boundary. Full history contains previous outcomes, so this result does not exploit time-homogeneous or low-dimensional Markov restrictions. In smaller models the efficient gradient can differ, and marginalized occupancy ratios may be more efficient. Claiming universal efficiency for (8) is incorrect.

6.3 Random slates: fixed design versus natural-generator target

When the generator is an externally fixed sampling protocol G_0 , (4) integrates over this known protocol in each $\widehat{V}_t$; (8) has no generator-score term. Exact integration is a mathematical assumption of these formulas. Finite Monte Carlo slate integration creates additional numerical error; use sufficiently accurate independent draws or carry that error into inference. One should not silently substitute a single logged slate into every integrated value and retain the same efficiency claim.

If instead the estimand keeps the **unknown natural** generator G_P from the observed population, then the target $d(P) = G_P\pi$ itself changes when P changes. In the full-history model the canonical gradient additionally contains

$$\sum_{t=1}^T \Omega_{t-1} \{W_t^{\pi, G_P}(H_t, C_t) - V_t^{\pi, G_P}(H_t)\}. \quad (9)$$

Derivation. A generator perturbation at t contributes its score $S_{G,t}(C_t|H_t)$ to the target g-formula. The remaining conditional target value after seeing C_t is $W_t(H_t, C_t)$. Change the measure only through the previous selection, since the generator distribution is unchanged between logging and target at this turn; this yields $E[\Omega_{t-1}(W_t - V_t)S_{G,t}]$. The conditional mean subtraction makes it a generator tangent component. Adding it to (8) gives the derivative for every model factor. $\square$

Equivalently, treat (H_t, C_t) as the decision state; the usual nonparametric longitudinal score then includes candidate variability. That score can also be used with a known generator as an unbiased but generally less efficient estimator. These are different observation models and efficiency statements, not conflicting algorithms. A fixed finite slate in the simulator makes the extra term identically zero.

7. History-specific causal Q and blip learning

The scalar value score does not by itself give a root-specific label for every prompt. For same-generator randomized slates, define the downstream corrected remaining-return variable

$$Z_{t+1} = \widehat{V}_{t+1}(H_{t+1}) + \sum_{s=t+1}^T \left(\prod_{k=t+1}^s \frac{\pi_k(J_k | H_k, C_k)}{\widehat{b}_k(J_k | H_k, C_k)} \right) \{R_s + \widehat{V}_{s+1}(H_{s+1}) - \widehat{Q}_s(H_s, C_s, J_s)\}, \quad (10)$$

with $Z_{T+1} = Y$. Integration over the fixed generator is understood in $\widehat{V}$. For each candidate index j in the realized slate **whose conditional action value is to be learned**, require both $b_t(j|h, c) > 0$ and $\widehat{b}_t(j|h, c) > 0$ at the relevant histories. This requirement also applies when $\pi_t(j|h, c) = 0$: target-policy support alone does not identify every candidate-specific Q. Excluded or zero-logging-probability candidates receive no such label. For each index meeting these conditions, define

$$\Gamma_{t,j} = \widehat{Q}_t(H_t, C_t, j) + \frac{1\{J_t = j\}}{\widehat{b}_t(j | H_t, C_t)} \{R_t + Z_{t+1} - \widehat{Q}_t(H_t, C_t, j)\}. \quad (11)$$

This is a vector of potentially noisy corrected labels over the supported indices of the finite realized slate. Only one residual is observed, so it is not a full-information counterfactual observation. Fit a critic by honest regression of these labels on (H_t, C_t, j) or on exact prompt text under justified exclusion; use cross-fitting at root level.

Theorem 6: exact conditional bias of the pseudo-outcome

For a queried index satisfying $b_j > 0$ and $\widehat{b}_j > 0$, put $z = (h, c)$, $b_j = b_t(j|z)$, $\widehat{b}_j = \widehat{b}_t(j|z)$, and

$$B_{t+1}(z, j) = E\{Z_{t+1} - V_{t+1}^d(H_{t+1}) \mid H_t = h, C_t = c, J_t = j\}.$$

Then

$$E(\Gamma_{t,j} \mid H_t = h, C_t = c) - Q_t^d(h, c, j) = \left(1 - \frac{b_j}{\widehat{b}_j}\right) \{\widehat{Q}_t(h, c, j) - Q_t^d(h, c, j)\} + \frac{b_j}{\widehat{b}_j} B_{t+1}(z, j). \quad (12)$$

Proof. Conditional on the slate and history, only $J_t = j$ contributes to the residual. Its expected multiplier is $b_j/\hat{b}_j$. By the Bellman equation, $E[R_t + Z_{t+1}|z, J = j] = Q_t^d(z, j) + B_{t+1}(z, j)$. Substitute in (11) and collect terms. $\square$

Consequently, local propensity correctness alone does not remove biased downstream labels. Conditional unbiasedness holds when the future recursion is conditionally unbiased at reachable next histories and the current propensity or current Q is correct. Theorem 4 applied starting from any such history supplies sufficient downstream conditions. Contrasts $\Gamma_{t,j} - \Gamma_{t,j'}$ estimate the conditional blip after regression under the same conditions.

Weighted squared loss, contrast losses, or ranking losses can train a neural critic, but they solve different statistical problems. Squared regression targets Q calibration; direct contrast loss can target heterogeneity; ranking loss targets decisions and need not calibrate Q. Bootstrap ensemble spread alone is not a proved pointwise confidence interval for the conditional causal effect. Smoothness, complexity, overlap, and distributional calibration assumptions are needed for honest conditional inference. With continuously varying histories or text, a point-evaluation Q is generally not a regular root- n functional in the unrestricted model; the root- n result for marginal policy value does not transfer to every neural prediction.

8. Optimality, policy improvement, and conservative selection

8.1 Supported optimal policy

For a fixed generator G , let $V_t^{*,G}$ solve

$$Q_t^{*,G}(h, c, j) = E\{R_t + V_{t+1}^{*,G}(H_{t+1}) \mid h, c, j\}, \quad V_t^{*,G}(h) = E_{C \sim G(\cdot|h)} \max_{j \in \mathcal{J}(h,C)} Q_t^{*,G}(h, C, j). \quad (13)$$

$\mathcal{J}$ contains eligible, supported candidates including STOP. With finite nonempty slates and measurable tie breaking, backward induction gives an optimal deterministic selector within this generator-constrained policy class. The maximization occurs after seeing the slate, so $E \max_j Q$ is appropriate; $\max_j E Q$ is generally not.

Proof. At the last decision, conditional expected utility is maximized by the largest supported Q. Assuming optimal continuation values are available at $t + 1$, the same pointwise maximization at t dominates every other first-stage selection followed by any admissible continuation. Integrating over the generator and inducting backward proves global optimality within the class. $\square$

Theorem 7: performance difference and certified local improvement

Let β and π be selectors sharing the same generator and environment. Then

$$\mathcal{V}(\pi) - \mathcal{V}(\beta) = \sum_{t=1}^T E_{\pi,G} \left[\sum_j \{ \pi_t(j \mid H_t, C_t) - \beta_t(j \mid H_t, C_t) \} Q_t^\beta(H_t, C_t, j) \right]. \quad (14)$$

If a simultaneous event guarantees $|\widehat{Q}_t^\beta(h, c, j) - Q_t^\beta(h, c, j)| \leq u_t(h, c, j)$ on all target-reachable histories/candidates, define

$$L_t(h, c; \pi, \beta) = \sum_j (\pi_j - \beta_j) \widehat{Q}_t^\beta(h, c, j) - \sum_j |\pi_j - \beta_j| u_t(h, c, j). \quad (15)$$

On that event, $\mathcal{V}(\pi) - \mathcal{V}(\beta) \geq \sum_t E_{\pi,G} L_t$. In particular, restricting policy changes to those with $L_t \geq 0$ pointwise guarantees nondecrease on the same event.

Proof. Under target trajectories sum the baseline Bellman residuals $R_t + V_{t+1}^\beta(H_{t+1}) - V_t^\beta(H_t)$. The state values telescope to $U^{tot} - V_1^\beta(H_1)$. Conditional averaging gives $E_G \sum_j \pi_j Q_t^\beta - V_t^\beta = E_G \sum_j (\pi_j - \beta_j) Q_t^\beta$, proving (14). Substituting $Q = \widehat{Q} + (Q - \widehat{Q})$ and bounding each error term below by $-|\pi_j - \beta_j| u_j$ proves (15) and its consequences. $\square$

The required simultaneous event is a substantial statistical condition. Ordinary per-candidate pointwise intervals, post hoc best-of-many selection, or uncalibrated neural uncertainty do not establish it. An average advantage on logging histories alone also

does not prove improvement under a changed trajectory distribution. A safer practical release gate is an independent root-level evaluation of the frozen proposed policy against baseline, with uncertainty for the policy-value difference.

STOP is handled by the same theorem. A deterministic continue candidate is preferred to STOP only if its supported downstream utility exceeds the stopping value after all future costs. The guarantee applies to the measured utility, not to an unmeasured preference or an oracle label leaked into policy input.

8.2 Finite-policy holdout guarantee

Suppose a training sample fixes m proposed policies and their nuisance functions. An independent evaluation set contains n independent roots, and a score for each policy lies in $[a, b]$ almost surely. If assignment probabilities are correct, its score mean is unbiased. Then, with probability at least $1 - \alpha$, simultaneously for every policy,

$$|\widehat{\mathcal{V}}(d) - \mathcal{V}(d)| \leq (b - a) \sqrt{\frac{\log(2m/\alpha)}{2n}}. \quad (16)$$

Proof. Apply Hoeffding’s inequality to each policy’s independent bounded root scores and union bound over m . If only a known bias bound r_d from (5) is available, add it to that policy’s radius. $\square$

A policy can be released when its lower value bound exceeds the baseline upper bound, or more efficiently when a paired score-difference lower bound exceeds zero. This is a finite-class holdout result; repeatedly inspecting the same holdout and generating additional policies violates its fixed-class premise. Fresh holdouts, a predeclared simultaneous class, or sequentially valid inference are needed for ongoing adaptation.

8.3 Candidate coverage versus critic error

To compare with unrestricted prompts, additionally assume slate exclusion, candidate-independent action reward/transition, a common cost convention, and an admissible prompt space $\mathcal{A}(h)$ with a measurable maximum or arbitrarily close maximizers. Let Q_t^*, V_t^* denote unrestricted optimal continuation values. Every slate maximum, greedy selection, and error bound in this subsection is over the **same eligible, positive-logging-support index set** $\mathcal{J}(h, c)$ used in (13), which is nonempty and contains STOP. Every compared selector assigns zero probability outside this set. Merely appearing in the slate does not make an excluded or unsupported candidate selectable. Define the candidate coverage loss

$$c_t(h) = V_t^*(h) - E_{C \sim G} \max_{j \in \mathcal{J}(h, C)} Q_t^*(h, C_j) \geq 0. \quad (17)$$

This is a theoretical quantity; the global optimal prompt over all language is usually unknown. It must not be estimated by labeling the best sampled candidate globally optimal.

Theorem 8: coverage decomposition and finite-horizon bounds

For any candidate selector $\hat{\pi}$ sharing G ,

$$\mathcal{V}^* - \mathcal{V}(\hat{\pi}, G) = \sum_t E_{\hat{\pi}, G} c_t(H_t) + \sum_t E_{\hat{\pi}, G} \left[\max_{j \in \mathcal{J}(H_t, C_t)} Q_t^*(H_t, C_{tj}) - \sum_{j \in \mathcal{J}(H_t, C_t)} \hat{\pi}_t(j | H_t, C_t) Q_t^*(H_t, C_{tj}) \right]. \quad (18)$$

If the critic estimates the **restricted optimal** $Q_t^{*,G}$ with uniform error at most ε_t over every relevant history, slate, and index in $\mathcal{J}(h, c)$, and $\hat{\pi}$ greedily selects its largest prediction **within that same set**, then

$$\mathcal{V}^* - \mathcal{V}(\hat{\pi}, G) \leq \sum_t \sup_h c_t(h) + 2 \sum_t \varepsilon_t. \quad (19)$$

Proof. Telescope unrestricted optimal state values along trajectories of $\hat{\pi}$: $\mathcal{V}^* - \mathcal{V}(\hat{\pi}) = \sum_t E[V_t^*(H_t) - Q_t^*(H_t, A_t)]$. Add and subtract the best eligible supported slate value; conditioning on pre-slate history yields (18). Every maximum is over $\mathcal{J}$, so an unselectable high-value candidate cannot eliminate the coverage loss.

For (19), first compare V_t^* with $V_t^{*,G}$. Their Bellman equations and $\max x - \max y \leq \max(x - y)$ imply

$$\sup_h (V_t^* - V_t^{*,G}) \leq \sup_h c_t(h) + \sup_h (V_{t+1}^* - V_{t+1}^{*,G}).$$

The terminal difference is zero, hence the unrestricted-versus-restricted gap is at most the sum of coverage suprema. Next, for any slate, let j^* maximize the true restricted optimal Q over $\mathcal{J}(h, c)$ and $\hat{j}$ maximize its estimate over that same set. Then

$$Q^{*,G}(j^*) - Q^{*,G}(\hat{j}) \leq |Q^{*,G}(j^*) - \widehat{Q}(j^*)| + |\widehat{Q}(\hat{j}) - Q^{*,G}(\hat{j})| \leq 2\varepsilon_t,$$

because $\widehat{Q}(j^*) \leq \widehat{Q}(\hat{j})$. Telescope $V^{*,G}$ along the greedy estimated policy; each conditional stage gap is at most $2\varepsilon_t$. Add the two bounds. $\square$

Do not apply (19) to a critic of an arbitrary baseline continuation. Such a critic supports Theorem 7’s improvement analysis, not a global near-optimality claim.

Coverage corollary. Suppose a protocol starts with K independent proposals conditional on history. Each proposal has probability at least $p_t(h)$ of simultaneously (i) having $Q_t^*(h, A) \geq V_t^*(h) - \eta_t$ and (ii) meeting a proposal-wise admissibility criterion that guarantees at least one corresponding index remains eligible and selectable with positive logging probability after the complete protocol. Assume $0 \leq \eta_t \leq B_t$, where B_t bounds the Q range. Then

$$c_t(h) \leq \eta_t + B_t \{1 - p_t(h)\}^K. \quad (20)$$

Proof. The qualifying near-optimal-proposal events are independent because they are functions of independent original draws and the fixed history. The probability of none is at most $(1 - p)^K$. If at least one occurs, the preservation condition puts an η_t -optimal index in the final selectable set $\mathcal{J}$, so the gap is at most η_t ; otherwise it is at most B_t . If r is the actual failure probability, the expected gap is at most $\eta_t + (B_t - \eta_t)r$. Since $0 \leq \eta_t \leq B_t$ and $r \leq (1 - p)^K$, this is at most $\eta_t[1 - (1 - p)^K] + B_t(1 - p)^K$, which implies (20). $\square$

Here K counts the **original independent draws**, not a random post-filter count or the final number of unique candidates. Deduplication that preserves one selectable copy of every qualifying proposal preserves the maximum event and does not invalidate the bound. Correlated beams, globally dependent eligibility filtering, or removing qualifying candidates can invalidate the independent-event or preservation premises; derive their coverage separately. An unsupported candidate has no qualifying probability for this corollary regardless of its oracle quality. More proposals can raise generation cost, so (20) for quality alone does not establish improved net utility as K grows.

9. Training a generative policy and a dynamic skill

An implementable cycle is: freeze generator and receiver; generate and randomize supported slates; fit Q/blip models; learn a selector on training roots; evaluate the frozen selector on new roots; only then use high-value evidence to update the generator and repeat collection. The learned controller is a dynamic skill only in the operational sense of an adaptive policy with a declared model/environment scope.

For a differentiable stochastic generator-selector law d_θ , with parameter-independent transition law and dominated differentiation, the policy-gradient identity is

$$\nabla_\theta \mathcal{V}(d_\theta) = E_{d_\theta} \sum_{t=1}^T \nabla_\theta \log d_{\theta t}(U_t | H_t) Q_t^{d_\theta}(H_t, U_t). \quad (21)$$

Derivation. Differentiate the trajectory density product to obtain a sum of action log-density derivatives multiplied by total return. For a stage’s score, rewards accumulated before that action have zero covariance with it because the conditional score

mean is zero. Conditional expectation of subsequent return given history and action is $Q_t^{d_\theta}$. Substitute to get (21). Any history-only baseline can be subtracted by the same zero-mean argument. $\square$

The identity uses on-policy history occupancy and the current policy’s true Q. Maximizing $E_{H \sim \text{old logs}, A \sim G_\theta} \widehat{Q}(H, A)$ is generally a surrogate; it omits changing history occupancy and can exploit critic error. Off-policy gradients require valid full likelihood ratios/occupancy correction and appropriate nuisance control. Distilling selected prompts, reward-weighted fine-tuning, direct preference objectives, and KL regularization are proposed optimization methods, not proved substitutes for (21) or randomized final evaluation.

A learned transition/world model can support planning only under identified transitions and sufficient state representation. Rollouts generated by that model are model predictions, not fresh counterfactual validation data. Longer simulated planning compounds transition error and can reward unrealistic states. Keep synthetic-world evaluation separate from actual frozen-receiver executions.

10. Honest inference, branch experiments, and transport

10.1 Shared prefixes and independent roots

A forked experiment restores the same declared receiver/tool snapshot and executes candidate continuations. Repeated rollouts estimate the conditional mean at that snapshot over receiver and continuation randomness. They do not reveal a unique user’s latent joint counterfactual outcomes. If shared random seeds are used, the coupling can reduce variance but is an experimental design choice; it is not the naturally occurring cross-world coupling.

All continuations sharing root task, common prefix, user, template, or stochastic cached output must stay in the same independence cluster when those shared variables create dependence relevant to the target. Fit nuisance models and select policies on other clusters. Reporting the total number of branches as n overstates independent information.

For m independent roots with a prespecified within-root averaging rule, let S_i be the root’s averaged evaluation score, including any valid branch-selection correction. The root-average estimand is estimated by $\widehat{V} = m^{-1} \sum_i S_i$, with

$$\widehat{\text{Var}}(\widehat{V}) = \frac{1}{m(m-1)} \sum_{i=1}^m (S_i - \bar{S})^2. \quad (22)$$

Justification. Independence applies to root scores, not branch scores. The sample variance of independent root scores is unbiased for their variance when roots are identically sampled; dividing it by m gives (22). A root-level central limit theorem or root bootstrap supports large-sample inference. Unequal/informative root sampling, adaptive branch allocation, or branch pruning needs explicit design weights; a convenience mean across all branches targets a branch-size-weighted distribution.

Branches must preserve the full system state. Restoring only visible text while tool files, caches, hidden conversation memory, or server model versions change violates the branch experiment’s consistency assumption. A behavioral simulator supplies exact counterfactuals only within its own specified simulator law.

10.2 Learned policies and nonregular optima

A conditional interval for a policy frozen after training is different from an interval for the population-optimal value. If different folds train different policies, the straightforward cross-fitted estimator targets an average of fold-trained policy values; it does not automatically evaluate the final all-data-trained policy. Evaluate that final policy on fresh roots. At ties or near ties, the maximum-value functional can be nonregular. Ordinary plug-in Wald intervals for an estimated argmax require extra conditions; the fixed-policy results above do not supply them.

Calibration reports should separately assess policy-value error, supported candidate ranking, blip sign, conditional mean calibration across predeclared history groups, uncertainty coverage, and realized deployment utility. A single global value interval does not establish individualized calibration.

10.3 New users and environment transport

A new user from the same target population is covered only to the extent that the relevant history is supported. For a different baseline-history distribution $q(H_1)$ with unchanged conditional mechanisms, known or estimable baseline density ratio $w = q/p$ can transport the value by weighting the initial distribution and scores. New receiver versions or different latent user preferences change conditional mechanisms and require stronger assumptions or new experiments. Logging version metadata enables auditing; it does not by itself prove invariance.

10.4 Historical logs versus prospective trials

Historical logs are useful for initialization, candidate proposal, and hypothesis development. Their causal use depends on measured confounders, treatment versions, stopping/missingness definitions, and support. A deliberately confounded simulation can demonstrate failure of a marginal reward model, but it should also include correctly adjusted sequential regression as a comparator. Prospective within-slate randomization removes selection confounding at that decision by design, provided the receiver continuation and outcome recording obey the protocol.

11. Research claims that remain conditional or open

1. **New free-form prompt performance:** not identified by the finite-slate randomization alone. Needs covered generation, structural generalization, or fresh trials.
2. **Uniform neural confidence bands:** assumed in Theorem 7, not constructed or validated here. Independent policy-level holdouts are the immediate defensible alternative.
3. **Universal representation sufficiency:** not established. Evaluate history/text compression through controlled ablations and conditional calibration; failure to detect bias does not prove sufficiency.
4. **Generator improvement after distillation:** requires new frozen-generator evaluation or valid full-generator ratios. Winner distillation can amplify noise and erase exploration.
5. **Prompt-level causal attribution to latent semantic factors:** requires explicitly defined semantic interventions and treatment-version assumptions. A latent vector coordinate is not automatically a manipulable treatment.
6. **Unmeasured confounding in human logs:** randomized selection evidence does not retroactively identify unsupported historical interventions. Sensitivity analysis can quantify assumptions but is not a proof they hold.
7. **Human personalization and preferences:** one frozen automatic evaluator estimates that evaluator’s outcome; it does not establish a user’s satisfaction, trust, or welfare.
8. **Joint receiver/generator adaptation:** outside the frozen-environment theorems unless adaptation state and intervention protocol are modeled. Changing both players is a different target.
9. **Multivariate win outcomes:** can be studied with a specified pairwise functional and coupled/independent policy draws, but do not inherit the scalar Bellman recursion without a defined scalar utility or augmented-state target.
10. **Novelty and practical superiority:** no theorem here establishes that the combined method is new or outperforms current prompt optimizers. Those are literature and empirical questions.

12. Source grounding

The proofs in this document are supplied explicitly for this project’s notation and assumptions; the underlying longitudinal DR and efficiency principles are established. Primary sources used to check those boundaries:

- Jiang, N. and Li, L. (2016). [Doubly Robust Off-policy Value Evaluation for Reinforcement Learning](#). Sequential DR evaluation and its policy-evaluation context.
- Bang, H. and Robins, J. M. (2005). [Doubly Robust Estimation in Missing Data and Causal Inference Models](#). Longitudinal and missing-data DR foundations; [author-hosted full text](#).
- Kallus, N. and Uehara, M. (2020). [Double Reinforcement Learning in Markov Decision Processes](#). Distinguishes efficiency models, non-Markov history ratios, and Markov structure. This project does not claim every history-weighted estimator is efficient under Markov restrictions.
- Luedtke, A. R. and van der Laan, M. J. (2016). [Statistical inference for the mean outcome under a possibly non-unique optimal treatment strategy](#). Nonregularity and inference for optimal versus learned policies.

- van der Laan, M. J. and Luedtke, A. R. (2015). [Targeted Learning of the Mean Outcome under an Optimal Dynamic Treatment Rule](#). Dynamic-rule learning and data-adaptive targets.

The attachment's recent paper titles, venue-status claims, and claims that a research area is unoccupied are research leads, not assumptions or validated novelty evidence for these theorems. The separate literature audit controls those statements.

Causal critic and generative policy: training specification

This is an implementation contract for the next experimental stage. The reference Python package validates finite-state estimation. It does **not** contain a trained neural prompt critic or fine-tuned prompt generator.

Critic inputs and outputs

Let $Z_t = (H_t, C_t)$ be the pre-selection history and realized slate. A critic $q_{\phi,t}(Z_t, j; \pi)$ predicts final quality or net utility after choosing index j and following a **named frozen continuation** π . The history encoder receives exact public conversation tokens and permitted public metadata. The action encoder receives the exact candidate string, STOP indicator, and permitted slate features. Use separate quality and cost heads and combine them using a declared utility conversion.

The encoder can share parameters with the generator only within the training partition. Freeze it during evaluation. A low-dimensional representation can improve statistical prediction, but full-history ignorability does not transfer automatically to the representation. Include a history-ablation/aliasing experiment. If the critic omits other candidates, state the slate-exclusion assumption and preserve enough context to define the action.

At a brand-new request, either (i) first run the fixed receiver and predict feedback values from the resulting history, which is the main design, or (ii) define an additional randomized initial-prompt rewrite decision. The original task is baseline, not a manipulable treatment unless the study explicitly defines a rewrite that preserves its target and scoring rules.

Backward learning for a fixed continuation

For notation here, treat the **realized slate as part of the observed state**, and evaluate a selector under the same natural candidate mechanism. All rewards and costs are encoded in terminal total utility U , so stage rewards are $R_t = 0$ in this recursion. Let $L_{T+1} = U$ and define $\hat{v}_t(z) = \sum_j \pi_t(j | z) \hat{q}_t(z, j)$. For each held-out fold, evaluate nuisance fits trained on other root tasks and recursively form

$$L_t = \hat{v}_t(Z_t) + \frac{\pi_t(J_t | Z_t)}{\hat{b}_t(J_t | Z_t)} \{L_{t+1} - \hat{q}_t(Z_t, J_t)\}.$$

The action-specific pseudo-outcome is

$$D_{t,j} = \hat{q}_t(Z_t, j) + \frac{\mathbf{1}(J_t = j)}{\hat{b}_t(j | Z_t)} \{L_{t+1} - \hat{q}_t(Z_t, J_t)\}.$$

Regress $D_{t,j}$ on (Z_t, j) to obtain a conditional-mean critic under the assumptions in the theory. When candidates are texts generated for each history, j alone has no cross-history semantic meaning; use exact candidate text and declared slate context as model inputs. The labels do not create information about an unseen candidate or an unseen semantic feature. Known experimental probabilities should be used exactly; estimating them unnecessarily can add error.

The theory distinguishes this natural-slate recursion from integrating over a fixed known proposal kernel. Use its corresponding score when claiming efficiency under the latter model. Candidate Monte Carlo integration error is not silently zero. All later continuation labels must be conditionally valid for $D_{t,j}$ to target the desired Q. Current-stage double robustness does not repair an arbitrary wrong continuation. Cross-fitting reduces dependence bias; it does not confer pointwise confidence coverage without conditional-regression assumptions.

Concrete fitting sequence

1. Partition complete root tasks into folds. Pin π , candidate-generator version, receiver, horizon, and utility.
2. Fit backward sequential Q on training folds using observed terminal outcomes and target-weighted next-state value. Use identical architectures to define the regression baseline.
3. Compute out-of-fold continuation and action labels with exact design probabilities. Retain extreme labels and report diagnostics; optional clipping is a separate biased sensitivity estimator.

4. Train a second-stage critic on labels. Evaluate its generalization on separate development tasks, and refit only within the training/development boundary after hyperparameters are locked.
5. Keep final testing independent of all fitting and model selection. A shared root prefix never crosses a fold or split.

For high-dimensional language, parameter sharing across stages and candidates is a modeling choice. It needs a fitted-model experiment, approximation-error analysis, and an out-of-task calibration check. The proofs do not assert neural convergence rates without a function class and regularity conditions.

Candidate selection and stopping

Given a slate, rank candidates by predicted net utility. Compare with STOP, whose value is the expected score of the current artifact minus costs already incurred; the hidden true score is unavailable at serving time. Include uncertainty/supportedness calibration before switching from a reference selector. A bootstrap spread from an arbitrary neural fit is not automatically a valid causal confidence bound.

A conservative deployment rule may switch only if a prespecified lower bound on average improvement in a calibrated subgroup is positive, or use a fixed threshold chosen on development tasks. Report it as a heuristic if no uniform error guarantee has been established. Theory bounds conditioned on a valid simultaneous error event are conditional guarantees, not a claim that an ensemble empirically supplies such an event.

Generator learning

The simplest generator update fits high-value feedback from training histories. For candidate advantages $\hat{A}_{tk} = \hat{q}_t(Z_t, k) - \sum_j \pi_{\text{ref}}(j | Z_t) \hat{q}_t(Z_t, j)$, an optional advantage-weighted objective is

$$\max_{\theta} \sum_{i,t,k} w_{itk} \log G_{\theta}(a_{itk} | H_{it}), \quad w_{itk} \propto \min\{w_{\max}, \exp(\hat{A}_{itk}/\beta)\}.$$

Normalize weights within history, tune β and $w_{\max}$ on development data, and preserve baseline candidates and exploration. This is a proposal-training heuristic, not an unbiased off-policy policy-gradient theorem. If proposal samples are used in a score-function update, record the proposal law and importance correction, and keep the critic fixed during each actor update. Any reference-policy KL term limits drift but cannot prove identification or prevent reward hacking by itself.

Train on continuation-correct values. Optimizing a one-step average of Q^{π} over historical states may be a policy-improvement surrogate; it is not the gradient of the complete new regime’s value without the proper occupancy distribution and policy-gradient conditions. Freeze each actor/selector version and recollect before making a policy-value claim. Changing both actor and continuation means retraining or re-evaluating the critic’s target.

Serving behavior

The virtual user receives an initial task, calls the frozen receiver, creates a candidate slate, estimates supported values, selects a prompt or STOP, and repeats until STOP or the horizon. It logs every transition. Values are always associated with receiver/generator/continuation versions. A receiver update invalidates an unqualified transfer claim. The system falls back to its declared reference policy when support or calibration checks fail.

This architecture supports prompt advice to a human and autonomous feedback as two deployment modes. Their outcomes need not agree: a human may edit the suggestion, supply private information, or ignore STOP. Evaluate human adherence and human-in-the-loop value separately if that becomes the intended application.

Experiment protocol v0.1

Status: prespecified development plan; confirmatory claims remain pending. The default budget is **\$0 paid API spend**. Local runs require an explicit call/time cap. Hardware-intensive training belongs to a separately budgeted issue. The root task/family, not each branch or turn, is the independent sampling unit.

Questions, estimands, and decision criteria

Study	Question	Primary target	Criterion / interpretation
S0 mathematical unit checks	Do weights, continuation, STOP, and robustness agree with exact truth?	Fixed-policy value in a finite DGP	Exact identities pass within numerical tolerance
S1 simulation	What protects inference from nuisance error, and when does overlap fail?	Bias/RMSE/coverage of value and contrasts	Report Monte Carlo uncertainty and failures; no selected best scenario
S2 fitted history–prompt critic	Can a learned critic rank interventions and calibrate conditional means?	Branch-specific mean value and prespecified subgroup effects	Out-of-task error/calibration/ranking versus strong sequential-regression baseline
S3 randomized LLM feedback	Do supported feedback choices have heterogeneous downstream effects?	Fixed-generator policy value and history-conditional contrasts	Prospectively collected known assignment; no hidden-answer inputs
S4 autonomous prompting	Does causal-critic selection improve quality or cost?	New-task policy value difference	Independent locked test; equal total inference budget; confidence interval
S5 generator updates	Does training the prompt generator add value beyond candidate ranking?	Value of each frozen joint generator/selector	Fresh randomization for each generator version; isolate generator and selector changes

S0–S1: known-truth simulation

Use a finite horizon with baseline task difficulty, evolving answer quality affected by prior prompts, history-dependent feedback assignment, and absorbing STOP. Exact enumeration/dynamic programming provides truth. The implemented reference suite documents its configuration and distinguishes oracle/fixed-nuisance experiments from fitted nuisance estimation.

Required full grid for the next agent: horizon 1/3/6; sample size 250/1,000/4,000 independent tasks; strong/weak overlap; observed versus deliberately omitted difficulty; equal and unequal branch counts; both nuisance models correct, only propensity correct, only all continuation-Q functions correct, and both wrong. Add history compression aliasing, changed candidate generator, hidden confounding, informative missingness, and generator overoptimization as failure studies. A failure setting is not an estimator defect when its identification assumptions were intentionally violated.

Compare unadjusted action groups, correctly specified sequential regression, misspecified sequential regression, ordinary IPW, longitudinal DR, clipped-weight sensitivity, and an oracle. Log bias, RMSE, empirical standard deviation, mean estimated standard error, 95% coverage, interval length, ESS by stage, maximum cumulative weight, zero-support rate, and stopping distribution. For 1,000 Monte Carlo replicates, nominal-coverage Monte Carlo SE is approximately 0.0069. Development runs may use fewer replicates, with their actual Monte Carlo uncertainty shown.

S2: fitted conditional value learning

Split root task families into 60% training, 20% development, and 20% final test; use a deterministic hash of family ID and a recorded seed. Cross-fit nuisances within training at the same root unit. Hyperparameters, prompt generators, stopping

thresholds, and utility conversions use training/development only. Encode only information available before selection. Every critic checkpoint names its continuation policy and generator.

Fit at least (a) a history-aware sequential outcome regression, (b) a longitudinal DR pseudo-outcome critic, and (c) a history-ablated associational critic. The third baseline diagnoses confounding but cannot be the only comparator. Use the same encoder, capacity, training data, and optimization budget for (a) and (b). Additional DRQ/DeepBlip adaptations must respect each method’s assumptions and compare using its actual published objective, not an arbitrary reimplementations with the name attached.

Report value mean-squared error, calibration slope/intercept and bins, candidate ranking regret, pairwise sign accuracy only for non-negligible true contrasts, and downstream policy value. Evaluate pointwise error only where repeated branches provide a usable conditional-mean reference; bootstrap entire root tasks when aggregating. Branch mean noise must be included in error analysis or reduced with independent replicate groups. Literal individual effects are unavailable even with repeated independent model seeds.

S3: randomized multi-round LLM interactions

Freeze a receiver by content digest and a manifest containing tokenizer, template, system message, quantization, temperature, sampling parameters, context limit and truncation rules, tool access, and service version. Freeze generator and evaluator separately. Use three feedback decisions after an initial answer, with a maximum context budget and explicit STOP.

Start with a slate of STOP, generic retry, verification, and a generated task-specific feedback prompt. These are experimental candidates, not a claim that language has four treatment classes. Later slates may contain four distinct generated texts plus STOP. The generator sees the current public conversation and public tests/constraints only. The primary information-parity comparison excludes hidden solutions, hidden test failures, and post-selection outputs. A separate oracle-feedback arm measures a different intervention and must be labeled.

For a four-candidate development slate, assign $b(j | h, c) = 0.25$ initially. An observed-history adaptive logger can initially use $0.8/K + 0.2 \text{softmax}(s(h, c))$, giving a .20 floor with four candidates. A separately declared stronger-imbalance stress arm may use $0.2/K + 0.8 \text{softmax}(s(h, c))$, whose floor is .05 at $K=4$. Publish the score function and full vector, not only the selected probability. Changes to the logger within a run require batch index in history and a suitable design analysis; the main study freezes it per batch.

A feasibility batch of 60 root tasks, one trajectory each, uses at most 7 model calls per task with one generator call and one receiver call per feedback decision plus one initial receiver call: 420 calls before STOP savings. Set a 60-minute cap and 1,024 new tokens per call. This batch chooses a competent benchmark and measures runtime/variance; it is not a test of superiority. A repeated-prefix calibration study starts with $80 \text{ held-out prefixes} \times 4 \text{ candidates} \times 4 \text{ independent continuations} = 1,280 \text{ branches}$. At a maximum of 3 remaining rounds, budget up to 8,240 receiver/generator calls ($1,280 \times 6 \text{ continuation calls plus } 80 \times 7 \text{ prefix calls}$), then replace this estimate with an exact plan before launching. Keep all branches of each root together.

The supplied `experiments/collect_ollama.py` is a bounded **collector smoke-test harness** for fixed slates of STOP/retry/verification/decomposition. It records exact requests, outputs, hashes, probabilities, token counts, failures, and a dry-run mode. It is not the trained generator, neural critic, or complete S3 study. Its arithmetic fixtures validate transport and parsing only. The handoff issue extends it with generator slates and benchmark adapters.

S4: locked prospective evaluation

Choose at most two primary policy comparisons before opening test outcomes: causal-critic adaptive selection versus history-aware fitted-regression selection, and versus a strong iterative-feedback baseline. For every policy, include initial receiver, generator, critic, reranking, and extra answer-sampling costs. Report quality at fixed total token/call budgets and the full quality–cost frontier. Giving one method hidden verifier feedback or extra calls creates a separate arm.

Baselines: initial answer/STOP; fixed generic retry; fixed verification; Self-Refine-style adaptive feedback; uniform/random candidate selection; history-aware fitted Q; DR critic; and a generated-slate best-of-K selector. OPRO/GEPA/PromptAgent are useful additional baselines if adapted faithfully to the same train/test split and total compute. They optimize different objects in some original settings, so label adaptations and include an unmodified reference where feasible. Do not require every literature method for a minimal first result.

Use task-matched execution of all frozen policies on independently restored environments, with several receiver seeds if affordable. Analyze one prespecified average per task, then paired task/family contrasts. A family bootstrap or cluster influence-function interval may be appropriate with enough independent clusters. Task IDs sampled from one benchmark do not establish generalization to all users or all LLMs.

Power is based on the root-task contrast D_i . For two-sided alpha .05 and 80% power, a planning approximation is $n \approx (1.96 + 0.84)^2 \sigma_D^2 / \delta^2$. Estimate variance on the feasibility sample and lock the test sample size before final outcomes. For binary paired outcomes with discordance .30 and a target difference .05, $\sigma_D^2 \approx .30 - .05^2$ gives approximately 933 tasks, not 60 or 100. With two prespecified comparisons, use Holm or conservative planning with alpha .025 (approximately 1,131 tasks under the same illustration). These are planning examples, not observed power or guaranteed coverage. Use task-family design effects or a cluster-based simulation when tasks within families are dependent.

S5: learned generative interventions

Train a generator on training histories using advantage-weighted text likelihood or a critic-guided score-function objective with a frozen critic and a reference-policy penalty. Clip/normalize training weights and tune on development data. Evaluate generator-only updates with a fixed selector and selector-only updates with a fixed generator, plus their combination. A generator can merely learn to exploit critic errors; report critic-predicted improvement against fresh-execution improvement and track diversity and unsupported-candidate rates.

Each new generator has a new version and collection batch. Candidate-selection propensities from the old generator do not evaluate its new language distribution. If a tractable joint slate-density ratio is unavailable, collect new randomized trajectories or use a prospective policy trial. A KL penalty does not create empirical support or guarantee causal calibration.

Stopping, failures, missingness, and transport

STOP preserves the current answer and incurs no later generation costs. Timeouts/service failures consume realized cost and yield a prespecified observable failure outcome for the deployment endpoint. A missing outcome is not converted to STOP. Record the reason, retain the assigned trajectory, and either use a justified missingness analysis or report worst/best bounds. Report both application robustness and quality conditional on successful service only if the latter is explicitly a secondary selected-population endpoint.

Separate receiver shift, task-distribution shift, evaluator shift, and generator shift. Reuse of a critic under a different receiver is a transport experiment requiring recalibration and fresh checks. Human logs add unmeasured intent, expertise, and external-information confounding; observational analyses must state these assumptions and include a sensitivity study rather than claiming that text encoders solve them.

Required release bundle

Return source revision; frozen manifests; dataset sources/licenses/splits; raw immutable JSONL or an access statement; checksums; deterministic collection and analysis commands; branch lineage; failures; total compute; estimator diagnostics; all pre-specified endpoints; and a claims ledger. Record independent reviewer findings before promoting a development result to validated evidence.

Synthetic reference experiment

Executed, CPU-only, \$0 external spend. No LLM or neural value model was trained or evaluated.

The target stochastic feedback policy has exact value **0.798750**. There are 400 independent Monte Carlo replications, each with 800 tasks and 2 continuations per task; horizon 3. Truth is computed by backward recursion and independently checked by complete trajectory enumeration.

H1 includes the initial receiver answer. At every active round the same finite candidate slate offers RETRY, CORRECT, and STOP. STOP freezes quality and incurs no further cost. The logged feedback depends on current answer quality, which earlier feedback can change. Terminal utility subtracts 0.035 per cost unit by default (RETRY costs 1, CORRECT costs 2).

Fixed-nuisance robustness and uncertainty

correct is the known simulator nuisance; wrong is a deliberately fixed misspecification. No nuisance fitting or cross-fitting performance is claimed. The code supplies task-fold assignment for future fitted experiments. All interval standard errors first average continuations within task.

Estimator	Exact bias	MC bias (MCSE)	RMSE	95% coverage (MCSE)
DR_both_correct	0.000000	0.00069 (0.00050)	0.00995	0.950 (0.011)
DR_propensity_correct_Q_wrong	0.000000	0.00075 (0.00052)	0.01036	0.953 (0.011)
DR_Q_correct_propensity_wrong	0.000000	-0.00377 (0.00279)	0.05595	0.950 (0.011)
DR_both_wrong	0.631259	0.62552 (0.00282)	0.62805	0.000 (0.000)
IPW_propensity_correct	-0.000000	0.00053 (0.00202)	0.04033	0.932 (0.013)
IPW_propensity_wrong	0.579312	0.57094 (0.00386)	0.57613	0.000 (0.000)
Direct_Q_correct	0.000000	0.00054 (0.00033)	0.00652	0.955 (0.010)
Direct_Q_wrong	-0.213881	-0.21301 (0.00054)	0.21328	0.000 (0.000)

The exact expectation calculation, not simulation alone, verifies double robustness. Coverage is finite-sample evidence for this toy process only. Zero estimated coverage MCSE at observed coverage 0 or 1 is a plug-in Monte Carlo value, not a certainty claim. `Direct_Q_correct` uses oracle conditional means and therefore has information unavailable to a fitted model. Intervals for biased estimators are shown to expose failure; they do not repair misspecification.

Confounding reversal

The following compares the next-answer quality after CORRECT versus RETRY. The causal contrast standardizes both interventions over the same active histories under the logging regime. It uses immediate quality (equivalently, terminal quality under STOP thereafter), not the main cost-penalized adaptive-policy outcome.

Stage	Observed contrast	Causal standardized contrast
1	-0.2694	0.2117
2	-0.2683	0.1680
3	-0.2598	0.1497

Exact policy values

Policy	Utility	Quality	Feedback calls
adaptive_stochastic	0.7987	0.8648	1.062
logging_behavior	0.6663	0.7802	2.264
always_retry	0.4106	0.5156	3.000

Policy	Utility	Quality	Feedback calls
always_correct	0.6555	0.8655	3.000
accept_initial	0.4750	0.4750	0.000
oracle_optimal	0.8683	0.9333	0.929

oracle_optimal is known-process dynamic programming, not a learned prompt generator. The conditional value file illustrates baseline-history-specific mean effects under two named continuation policies; these are not realized individual treatment effects.

Reproduce and extend

```
uv run \
  --extra dev python scripts/run_synthetic.py \
  --replicates 400 \
  --tasks 800 \
  --episodes-per-task 2 \
  --seed 20260919 \
  --horizon 3 \
  --cost-penalty 0.035 \
  --output results/new_reproduction
uv run --extra dev pytest -q
```

Use a fresh output path; the runner refuses to overwrite an existing result. Config records seeds, source hashes, costs, sampling unit, and estimator definitions. Replicate estimates, exact policy values, conditional effects, overlap moments, and environment details are separate machine-readable files.

Limitations: finite supported candidates, randomized known behavior, four baseline strata, a fixed receiver transition law, no hidden confounding, no generator shift, no representation loss, no learned policies, no human users, and no empirical LLM effectiveness claim. Real experiments require frozen checkpoints, task-level train/calibration/test splits, fresh randomized continuations, terminal scorer isolation, failure accounting, and predeclared cost/support diagnostics.

Handoff to the experimental agent

The user authorized development of the theory and experiment package and communication through this GitHub repository. The next agent should claim the corresponding issue, implement on a branch, and return a reviewable PR. Posting the issue is not evidence that any agent has accepted or executed it.

Read first

AGENTS.md, docs/research_proposal.md, docs/theory.md, docs/training_and_serving.md, docs/experiment_protocol.md, docs/data_contract.md, and docs/literature_audit.md define the scientific contract. Start by independently reproducing the reference suite and inspecting failures. Do not transplant numerical results from DTR-AgentEvals: that is a different intervention study.

Work packages

Priority	Package	Deliverable	Completion criterion
P0	Independent theory/code audit and fitted-nuisance simulation	Proof-review notes; exact-truth checks; task-cross-fitted nuisance learner; complete robustness grid	No unresolved identification/score mismatch; actual fitted-model bias/coverage reported with MCSE
P1	Randomized generated-feedback dataset	Open-weight collector; restored-state branches; immutable logs; manifests; failure report	Exact slates/probabilities; no answer leakage; budget adherence; valid split/cluster structure
P2	Critic, ranking, and STOP	Matched regression and DR critics; conditional-mean calibration; ranking regret	Held-out tasks; named continuation; unsupported predictions flagged; all costs counted
P3	Learned generator and locked prospective trial	Generator/selector ablations; frozen policy comparison; uncertainty and compute ledger	Fresh evaluation after generator changes; independent test; all prespecified comparisons reported

P0 can begin on CPU. P1 can run a small feasibility batch with an available local receiver after its manifest is pinned. P2 depends on an adequate dataset, and P3 depends on a successful leakage/identification audit. If hardware is unavailable, complete CPU work and report the exact pending resource need; do not label a dry run as an LLM experiment.

Reply format on the issue

State the claim and scope, branch and commit, files changed, commands, test status, dataset/model digests, actual calls/tokens/runtime/spend, failures and missing outcomes, statistical results with uncertainty, and which claims remain unsupported. Attach tables/raw-result paths and explain any protocol deviation before selecting a final policy. Request review of a concrete PR rather than silently merging changes into the baseline.

Non-negotiable comparisons

The causal critic must compete with a correctly history-conditioned sequential-regression critic using the same encoder and tuning budget. Include a simple iterative-feedback baseline and account for generator cost. A confounded marginal comparison is a diagnostic, not the strongest baseline. Finite-action simulations validate statistical identities; they do not validate a neural treatment embedding or a free-form generative policy.

Ownership and acceptance

The initial package owns the theory specification and reference code. The experimental agent owns implementation and experiment execution on its branch after claiming a work package. Reviewer acceptance is distinct from execution completion. Keep issue/PR links in docs/project_status.md; never imply another agent has started solely because an issue exists.